

# DWH2 Final Projekt

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<https://github.com/bjoern555/dwh2-text-to-sql>



# Dataset Choice & Rationale

| Difficulty                                                                                   | Count     | Question IDs                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Easy        | 3         | 779, 268, 297                                                                                                                                                                                                                                                                                                                                                            |
| Medium      | 6         | 89, 506, 708, 569, 861, 605                                                                                                                                                                                                                                                                                                                                              |
| Hard        | 5         | 94, 155, 259, 426, 534                                                                                                                                                                                                                                                                                                                                                   |
| Extra-Hard  | 4         | 120, 284, 551, 591                                                                                                                                                                                                                                                                                                                                                       |
| <b>Total</b>                                                                                 | <b>18</b> | <b>9     9   =&gt; 50/50</b> |

## Why FootballDB?

- Universally familiar domain
- Query results can be validated intuitively
- No need to study the database schema in depth

# System Architecture / Setup

## LLM

- Local Ollama with *llama3.2:1b* or *qwen2.5-coder:14b*
- Configurable via .env (OLLAMA\_MODEL)

## Database

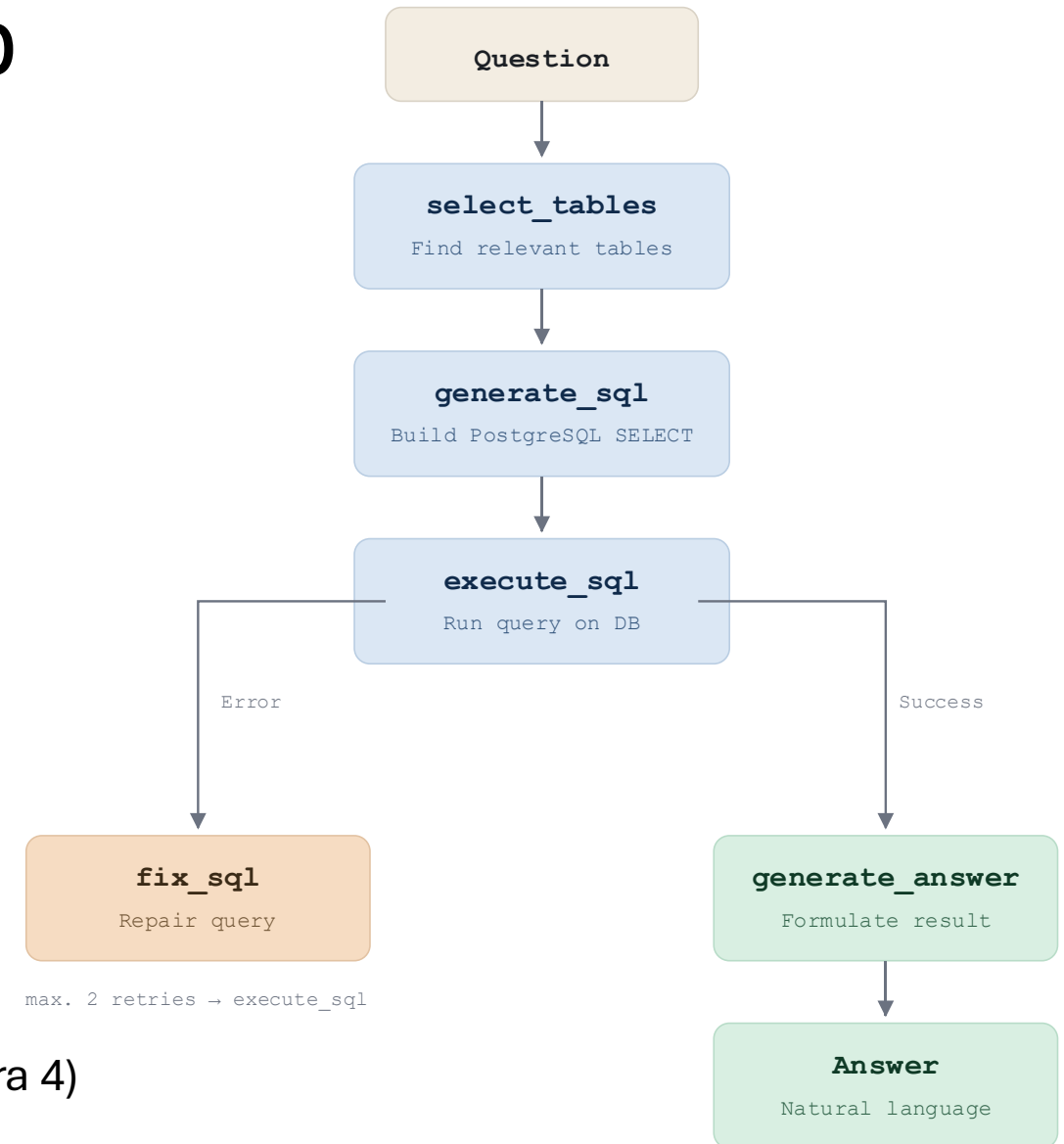
- PostgreSQL, Schema exp\_v3, hosted on ZHAW-Server
- Connection via LangChain

## Approach

- LangGraph-based multi-step reasoning

## Evaluation Metric

- Execution Accuracy: Predicted result == Gold result (exact match)
- Test Set: 18 Questions (Easy 3 / Medium 6 / Hard 5 / Extra 4)



# Query Examples - 1/3 correct

Case 1

WRONG

id 779 · easy

*"Where was the world cup in 2018?"*

Gold: `SELECT venue FROM world_cup WHERE year = 2018`

Pred: `SELECT T1.venue FROM world_cup T1 INNER JOIN stadium T2 ON T1.venue = T2.city WHERE ...`

Gold result: `[('Russia',)]`      Pred result: empty

Issue: Unnecessary join on stadium table

Case 2

CORRECT

id 259 · hard

*"Show all national teams who have won more than 2 world cups"*

Gold: `... JOIN national_team T2 ... WHERE winner = 'true' GROUP BY teamname HAVING COUNT(year) > 2`

Pred: `... INNER JOIN world_cup_result T2 ... WHERE winner = 'TRUE' GROUP BY teamname HAVING ...`

Both results: `[Argentina, Italy, Brazil, Germany]`

Note: Different formulation, identical result

Case 3

WRONG

id 534 · hard

*"Which players played for Switzerland in 2014? Return the names of the players"*

Gold: `... JOIN player_fact ... JOIN national_team ... WHERE teamname = 'Switzerland' AND year = 2014`

Pred: `... JOIN player_club_team T2 ... WHERE T2.club_id = 'Q2701659' AND T1.country = 'Switzerland' ...`

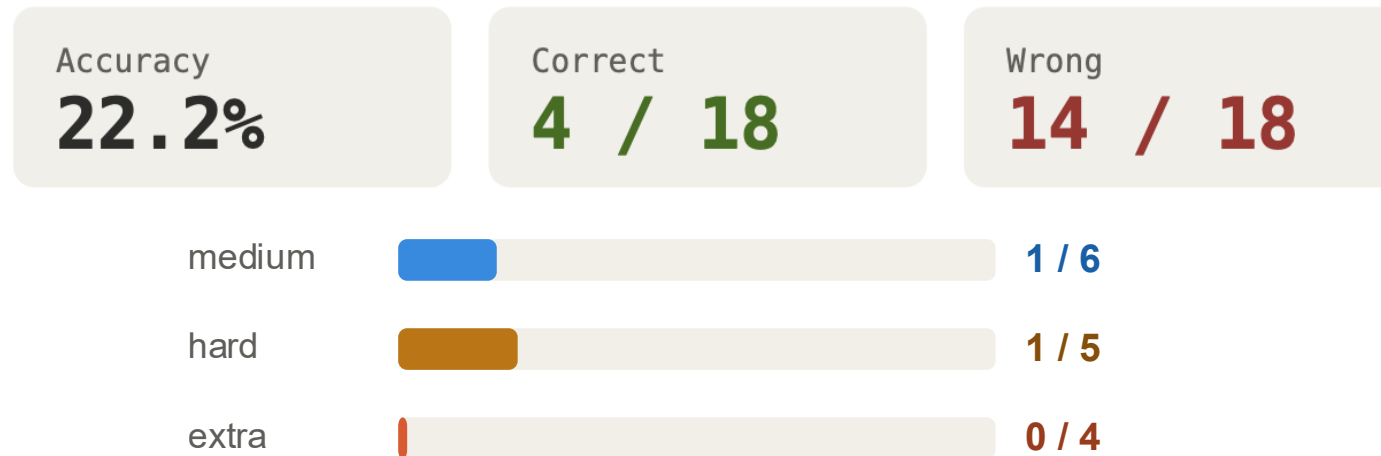
Gold result: 23 players · Pred result: SQL error

Issue: Hallucinated column `player.country`

# Execution Accuracy Results

*First baseline run. Weak overall, but a starting point we improve on later*

- Accuracy: 22.2 % with [the llama3.2:1b model](#)



- The model handles trivial lookups but collapses on anything requiring joins or multi-step reasoning (0/4 on extra-hard)

# Analysis of Errors / Model Limitations

- The *llama3.2:1b model* fails almost entirely at the schema level, **wrong joins** and **hallucinated columns**

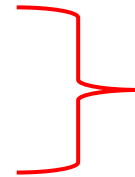
## Most common failure causes (14 wrong)

**6×** Wrong join (incorrect table linked)

**5×** Hallucinated / invalid column

**2×** Queried the wrong table

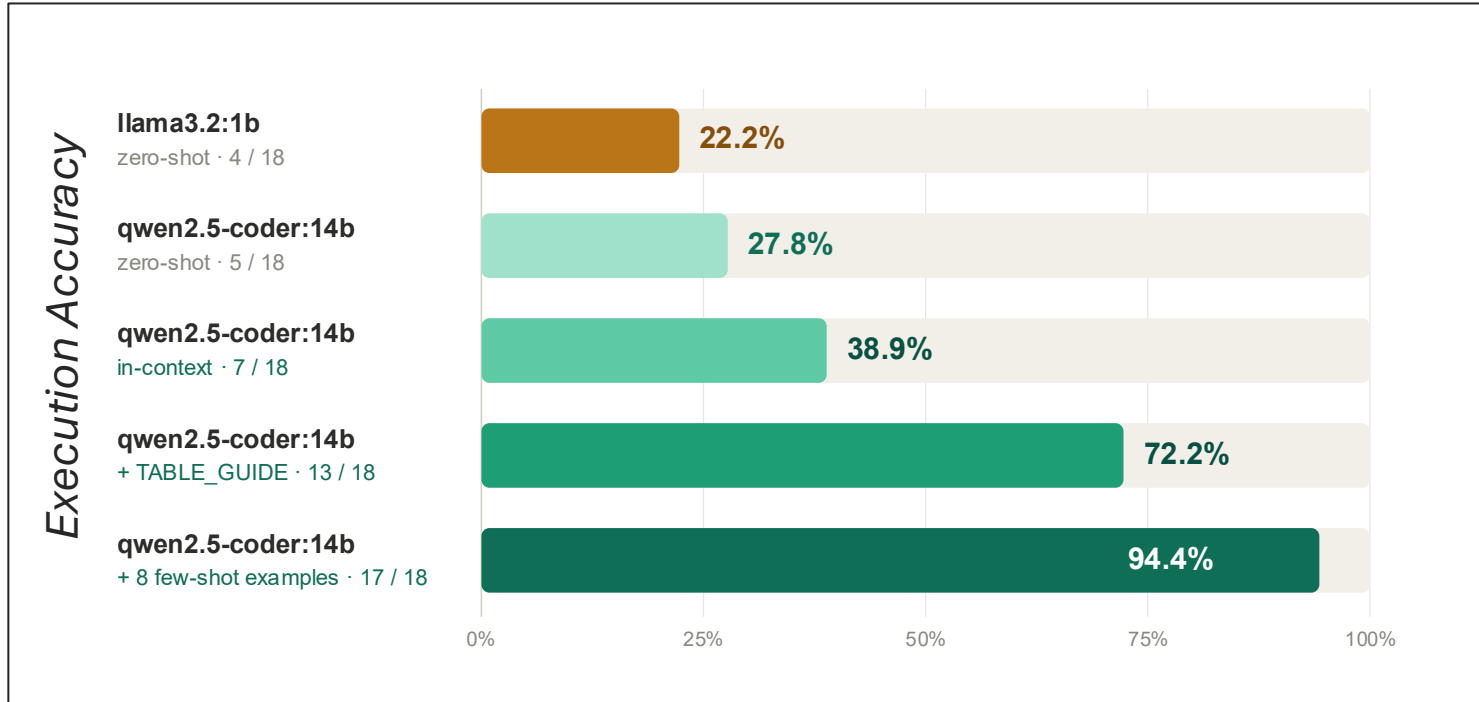
**1×** Missing filter / wrong syntax



*This indicates that we need a TABLE\_GUIDE*

- This tells us that *llama3.2:1b model* is too weak and thus we need a better model that doesn't hallucinate
- In-context learning / few-shot examples could help

# Optional Techniques Implemented



- The biggest win was adding the TABLE\_GUIDE to the llm model
- In combination with a stronger model, this resulted in an accuracy of 72.2%
- Adding few-shot examples helped but endangered the model to overfit on training set
- At the end we achieved a success rate of 94.4%, while the last error was only an additional column

# Conclusion



We managed to adjust our model to reach a successrate of 94% with a start of 22%



Simple LangGraph-based multi-step reasoning is only working with additional enhancements



The strongest impact was from describing the tables to the model and adjusting the model from llama to qwen



Today is a good day to get 20/20 points ;).. because we worked soo hard

# Future Work

**Model:** qwen2.5-coder:14b + in-context + TABLE\_GUIDE + few-shot  
18 additional validation queries

Accuracy

**50.0%**

Correct

**9 / 18**

Wrong

**9 / 18**

*We created a new data set for testing if our model overfits or generalizes*

- Results improved, as expected -> from 22.2% to 50% accuracy
- But they did not reach the success rate of our original dataset (94.4%)
- It scored more than 70 % of the easy tasks, none of the harder once
- Conclusion: the system does not generalize perfectly, it is partly overfitting to our dataset.

## Most common failure causes (9 wrong)

**4x**



**Wrong aggregation / counting logic**  
e.g., SUM, COUNT, GROUP BY mistakes

**2x**



**Incorrect joins causing duplicate rows**  
e.g., missing or wrong join conditions

**2x**



**Missing or incomplete results**  
e.g., filters not applied, partial outputs

**1x**



**Wrong interpretation of match outcome**  
e.g., winner vs opponent confusion